Carbon tetrachloride:

1. is used as dry cleaning agent
2. is used as a solvent for oils, fats and waxes
3. As the vapour of CCl_4 is non – combustible, it is used under the name pyrene for extinguishing the fire in oil or petrol.

14.4 Organo metallic Compounds

Organo metallic compounds are organic compounds in which there is a direct carbon –metal bond. For Example

$CH_3Mg\,I$ - Methyl magnesium iodide

CH_3CH_2MgBr - Ethyl magnesium bromide

The Carbon - Magnesium bond in Grignard reagent is covalent but highly polar. The carbon atom is more electro negative than magnesium. Hence, the carbon atom has partial negative charge and the magnesium atom has partial positive charge

$$R^{\delta-} \text{ ----- } Mg^{\delta+}X$$

14.4.1. Preparation

When a solution of alkyl halide in ether is allowed to stand over pieces of magnesium metal, the metal gradually dissolves and alkyl magnesium halide (Grignard reagent) is formed. All the reagents used should be pure and dry

Example

$$\underset{\text{Iodomethane}}{CH_3{-}I} + Mg \xrightarrow{\text{dry ether}} \underset{\text{Methyl magnesium iodide}}{CH_3MgI}$$

Evaluate Yourself

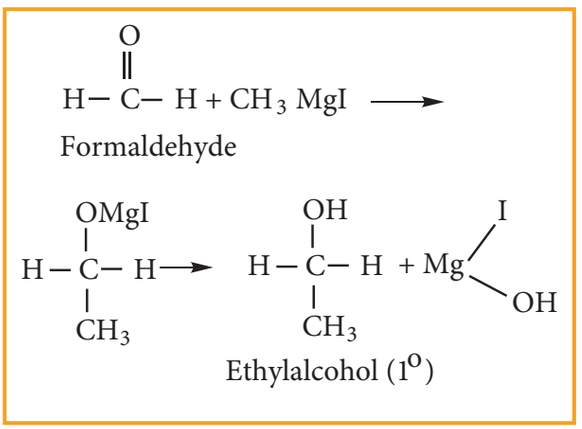
5) Why Grignard reagent should be prepared in anhydrous condition?

14.4.2. Uses of Grignard reagent

Grignard reagents are synthetically very useful compounds. These reagents are converted to various organic compounds like alcohols, carboxylic acids, aldehydes and ketones. The alkyl group being electron rich acts as a carbanion or a nucleophile. They would attack polarized molecules at a point of low electron density. The following reactions illustrate the synthetic uses of Grignard reagent.

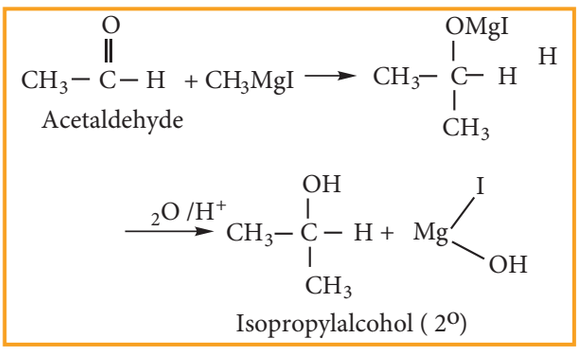
(1) Preparation of primary alcohol

Formaldehyde reacts with Grignard reagent to give addition products which on hydrolysis yields primary alcohol.



2) Preparation of secondary alcohol

Aldehydes other than formaldehyde, react with Grignard reagent to give addition product which on hydrolysis yields secondary alcohol.



3) Preparation of Tertiary alcohol

Ketone reacts with Grignard reagent to give an addition product which on hydrolysis yields tertiary alcohols.

Unit 14.indd 239 23-12-2021 17:22:51